

Braedyn Thompson

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EDUCATION

University of California, Berkeley

Graduation: Fall 2027

Major: Computer Science & Data Science - Applied Mathematics & Modeling Emphasis

GPA: 3.75/4.00

Coursework: Artificial Intelligence, Data Structures, Principles of Data Science, Data Engineering, Data Mining and Analytics, Introduction to Machine Learning

WORK & PROFESSIONAL EXPERIENCE

Lawrence Berkeley National Laboratory

Berkeley, CA

Data Science & Machine Learning Research Intern

Sep 2026 – Present

- Implemented PatchTST (PyTorch/neuralforecast) for multivariate time-series forecasting of cache-traffic signals across multiple horizons, tuning lookback windows and beating a persistence baseline on all targets by up to 24% RMSE.
- Exposed the ceiling on peak-load predictability and guided feature/model design by building reproducible Pandas EDA pipelines over 3 years of XCache traffic logs, visualizing seasonality, change-points, and right-skewed peaks with Matplotlib, Seaborn, and Plotly, and using rolling cross-validation to show models under-predict rare peaks.

California Baptist University

Riverside, CA

Machine Learning Research Intern

May 2026 – Present

- Diagnosed the root cause of a stalled inverse-FEA predictor by benchmarking 7 ML models (Random Forest, XGBoost, HGB/ExtraTrees ensembles, PCA pipelines) and running paired clean-vs-noisy signal analysis, proving the $R^2 \approx 0.10$ ceiling was a data limitation rather than a model limitation and redirecting the team from futile model tuning.
- Raised earthquake-detection accuracy to 0.992 ROC-AUC (vs. 0.550 baseline) and magnitude-estimation R^2 to 0.840 by training PyTorch CNN→Transformer and GNN ensembles on 5,800+ multi-station SCEDC waveform windows.
- Built a real-time daemon streaming 10 SeedLink stations that runs continuous PyTorch CNN→Transformer detection plus GNN-fused multi-station magnitude estimation, confirming events via graded multi-station coincidence and geographic move-out checks before dispatching FCM push alerts, deployed on an Oracle Cloud VM under systemd.

NumIsToken

Berkeley, CA

Full-Stack Software Engineering Intern

Sep 2025 – Jun 2026

- Enabled resumable redemptions for previously unrecoverable transactions by building Redemption History and Detail services with robust state management and payment resumption logic, eliminating a class of stuck-transaction failures.
- Reduced redemption action load times by ~85% (~2–3s to ~300ms) by removing redundant function calls to streamline backend lifecycle APIs, then containerizing and deploying backend services with Docker for production readiness.

PROJECTS

Academy of Testers

Berkeley, CA

Machine Learning & Artificial Intelligence Software Developer

Jan 2023 – Jun 2026

- Architected an adaptive learning engine that models per-student skill mastery by combining Bayesian Knowledge Tracing with Item Response Theory, selecting questions to maximize information gain at learner's estimated ability, and extended the 8-skill model with forgetting-curve decay and prerequisite-graph penalty propagation, powering a mastery radar.
- Improved AP essay-grading consistency, as measured by a ~23% reduction in scoring error (QWK 0.62) against official College Board scores, by building a RAG pipeline that embeds each free response, retrieves rubric clauses and score banded exemplars from a Postgres vector store using cosine similarity, and returns calibrated results as constrained JSON.

BearLM

Berkeley, CA

Data & Artificial Intelligence Engineer

Aug 2026 – Present

- Built a fully local, zero-cost RAG Q&A assistant (LangChain, Ollama, Llama 3.1 8B, nomic-embed-text, Chroma) over ~10K chunks spanning UC Berkeley CS/DS courses; blended keyword and meaning-based search (BM25 + vector embeddings) with reranking to pull the most relevant sources, lifting first-result accuracy from 56% to 82% (top-5 86%).
- Reduced hallucinated answers by grounding every response strictly in its retrieved sources, combining a relevance filter that scores and drops weak matches, query rewriting for sharper retrieval query, strict prompting answers only from context, and clickable citations page-level citations; raising RAGAS faithfulness from 0.55 to 0.83 (0.97 relevance).
- Built the data ingestion pipeline converting mixed-format materials (PDF, md, ipynb) into token-based, metadata-tagged embeddings; added analytics that mine query logs to surface the most-asked topics and auto-generate practice questions.

TECHNICAL SKILLS

Python, SQL, Pandas, NumPy, Matplotlib, Seaborn, Plotly, PyTorch, scikit-learn, Neural Networks, CNN, GNN, Transformers, RAG, Reranking, LangChain, Ollama, Chroma, RAGAS, Prompt Engineering, PostgreSQL, MySQL, SQLite, Docker, Git

Lawrence Berkeley National Laboratory, Data Science & Machine Learning Research Intern

At Lawrence Berkeley National Laboratory, I work with large datasets to develop and evaluate machine learning models for time series forecasting. I implemented PatchTST using PyTorch and NeuralForecast to forecast multivariate cache traffic across multiple horizons, tuning model parameters and achieving up to a 24% improvement in RMSE over the persistence baseline. I also built reproducible Pandas data analysis pipelines over three years of XCache traffic logs, analyzing seasonality, change points, and traffic peaks using Matplotlib, Seaborn, and Plotly. Through rolling cross validation, I evaluated model performance and identified patterns in model behavior, including consistent underprediction of rare traffic peaks.

California Baptist University, Machine Learning Research Intern

At California Baptist University, I analyze large scientific datasets and develop machine learning solutions for biomedical and seismic applications. I benchmarked seven machine learning models for an inverse FEA research project and analyzed their performance to determine that an R^2 ceiling of approximately 0.10 was caused by limitations in the underlying data rather than model selection. I also developed a PyTorch CNN and Transformer system combined with GNN ensembles for earthquake detection using more than 5,800 SCEDC waveform windows, achieving a 0.992 ROC AUC compared with a 0.550 baseline. I deployed the resulting system to process data from 10 SeedLink stations and used multistation analysis to identify events before sending alerts through Oracle Cloud.

NumIsToken, Full Stack Software Engineer Intern

At NumIsToken, I developed backend services and data workflows for a digital coin collecting platform. I worked on redemption history, redemption detail, state management, and payment resumption logic while analyzing application behavior to identify and resolve transaction failures. I reduced redemption action load times from approximately two to three seconds to around 300 milliseconds by identifying and removing redundant function calls. I also containerized and deployed backend services using Docker while working across APIs, application logic, and database interactions.

The Kigumi Group, Backend Software Engineer Intern

At The Kigumi Group, I worked on backend systems for an AI educational platform and focused on improving system reliability through testing and data driven debugging. I increased code coverage from 90% to 98% through unit and integration testing and investigated production issues affecting AI assistant functionality. I identified problems involving unstable backend connections and malformed query formulation for image generation and GPT models, then implemented targeted fixes to improve the reliability of the system and its AI powered workflows.

Academy of Testers, ML & AI Software Developer

At Academy of Testers, I developed data driven machine learning and AI systems for an adaptive learning platform. I architected an adaptive learning engine using Bayesian Knowledge Tracing and Item Response Theory, incorporating information gain, forgetting curve decay, and prerequisite relationships to personalize learning recommendations. I also built a retrieval augmented generation pipeline for AP essay grading that embedded student responses, retrieved relevant rubric clauses and score banded exemplars from a vector store, and generated structured outputs for evaluation. The system reduced essay grading inconsistency by approximately 23% while using data and model outputs to improve the learning experience.

BearLM, Data & Artificial Intelligence Engineer

At BearLM, I built a local AI question answering system over approximately 10,000 chunks of UC Berkeley Computer Science and Data Science course materials. I developed an ingestion and analysis pipeline for PDFs, Markdown files, and notebooks, storing document metadata and embeddings for retrieval. I combined BM25 search, vector embeddings, reranking, and relevance filtering to improve retrieval accuracy, increasing recall at one from 56% to 82% and achieving 86% recall at five. I also evaluated system performance using RAGAS and improved faithfulness from 0.55 to 0.83 through query rewriting, relevance filtering, and stricter grounding.